MTH 204 : Lecture 13

Basic Theory :

In general an ODE system is of the form:

$$\left.\begin{aligned} y_1' &= f_1(t, y_1, \dots, y_n) \\ y_2' &= f_2(t, y_1, \dots, y_n) \\ &\vdots \\ y_n' &= f_n(t, y_1, \dots, y_n) \end{aligned}\right\}$$

We can write the system as a vector equation by $Y' = f(t, Y)$ where $Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ and $f = \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{pmatrix}$

An initial value problem needs n initial conditions at t_0 i.e. $y_1(t_0) = K_1,\ y_2(t_0) = K_2, \dots, y_n(t_0) = K_n$

This can also be written as

$$Y(t_0) = K \quad \text{where} \quad K = \begin{pmatrix} K_1 \\ K_2 \\ \vdots \\ K_n \end{pmatrix}$$

$$\begin{array}{c} \| \\ \begin{pmatrix} y_1(t_0) \\ y_2(t_0) \\ \vdots \\ y_n(t_0) \end{pmatrix} \end{array}$$

Existence and Uniqueness:

Let $f_1, f_2, \ldots, f_n$ be continuous functions with continuous partial derivatives $\frac{\partial f_1}{\partial y_1}, \frac{\partial f_1}{\partial y_2}, \ldots, \frac{\partial f_n}{\partial y_n}$ in some domain R

of the $t y_1 y_2 \ldots y_n$ space containing the point $(t_0, K_1, \ldots, K_n)$

Then the ODE system has a solution on some interval $t_0 - \alpha < t < t_0 + \alpha$ satisfying the initial conditions, and this solution is unique.

Linear Systems:

$$\left.\begin{aligned}
y_1' &= a_{11}(t)y_1 + a_{12}(t)y_2 + \cdots + a_{1n}(t)y_n + g_1(t) \\
y_2' &= a_{21}(t)y_1 + a_{22}(t)y_2 + \cdots + a_{2n}(t)y_n + g_2(t) \\
&\vdots \\
y_n' &= a_{n1}(t)y_1 + a_{n2}(t)y_2 + \cdots + a_{nn}(t)y_n + g_n(t)
\end{aligned}\right\}$$

The above can be written as $\boxed{Y' = AY + g}$

where $Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ and $g = \begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{pmatrix}$

and $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$

If $g = 0$, the system is homogeneous.
(i.e. $Y' = AY$)

Existence and Uniqueness:

Let the a_{ij}'s and g_i's functions be continuous functions of t in an open interval I containing t_0, then there exists a solution, satisfying the initial conditions and this solution is unique.

Superposition Principle:

The linear combination of any two solutions Y_1 and Y_2 of the Homogeneous Equation is also a solution of the Homogeneous Equation.

Proof: Let $Y = c_1 Y_1 + c_2 Y_2$

Now, $Y' = (c_1 Y_1 + c_2 Y_2)' = c_1 Y_1' + c_2 Y_2'$

$= c_1 AY_1 + c_2 AY_2 = A(c_1 Y_1 + c_2 Y_2) = AY$

General Solution : Wronskian :

If the a_{ij}'s functions are continuous, then the general solution of the Homogeneous equation can be written as :

$$Y = c_1 Y_1 + c_2 Y_2 + \cdots\cdots + c_n Y_n$$

where $Y_1, Y_2, \ldots, Y_n$ constitute a basis or fundamental system of solutions and there is no singular solution.

We can write the n basis functions as columns of a matrix $\tilde{Y}$

$$\tilde{Y} = (Y_1, Y_2, \ldots, Y_n) = \begin{pmatrix} Y_1^{(1)} & Y_2^{(1)} & \cdots & Y_n^{(1)} \\ Y_1^{(2)} & Y_2^{(2)} & & Y_n^{(2)} \\ \vdots & \vdots & & \vdots \\ Y_1^{(n)} & Y_2^{(n)} & & Y_n^{(n)} \end{pmatrix}$$

and write the general solution as ;

$$Y = \tilde{Y} c \quad \text{where} \quad c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

$$\left(Y = c_1 Y_1 + c_2 Y_2 + \cdots + c_n Y_n = [Y_1 \; Y_2 \; \cdots \; Y_n] \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \right)$$

The Wronskian W is the determinant of $\tilde{Y}$

$$\text{i.e.} \quad W = |\tilde{Y}|$$

Note: The solutions form a basis on the interval iff $W \neq 0$ at some point on the interval. W is either identically zero or nowhere zero on the interval.

Constant Coefficient System :

Consider $Y' = AY$ where A is a $(n \times n)$ matrix of constants.

We try a function of the form $Y = Xe^{\lambda t}$

$$\Rightarrow Y' = \lambda X e^{\lambda t}$$

$$\Rightarrow \lambda X e^{\lambda t} = A X e^{\lambda t}$$

$$\Rightarrow AX = \lambda X$$

Note that if $Y = \begin{pmatrix} e^{\lambda t} x_1 \\ \vdots \\ e^{\lambda t} x_n \end{pmatrix}$

then $Y' = \begin{pmatrix} \lambda e^{\lambda t} x_1 \\ \vdots \\ \lambda e^{\lambda t} x_n \end{pmatrix}$

Thus Y is a solution of the system of ODE if X is an eigenvector of A.

If A has n distinct eigen values, then the general solution is

$$Y = c_1 X_1 e^{\lambda_1 t} + \cdots\cdots + c_n X_n e^{\lambda_n t}$$

The Wronskian of the basis of solutions is

$$W = \left| X_1 e^{\lambda_1 t} \cdots\cdots X_n e^{\lambda_n t} \right|_{n \times n}$$

$$= e^{\lambda_1 t + \cdots\cdots + \lambda_n t} \left| X_1 \cdots\cdots X_n \right|_{n \times n}$$

- The exponential term cannot be zero and the determinant of the matrix of eigenvectors cannot be zero because they are linearly independent vectors since they correspond to distinct eigenvalues. This proves that there is no singular solution if all eigenvalues are distinct.

(It is true whenever the n-solutions are linearly independent)

Phase Plane Method: (Qualitative Method):

We will look at systems with constant coefficients consisting of two ODEs

$$Y' = AY \quad \Rightarrow \quad \left.\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 \\ y_2' &= a_{21}y_1 + a_{22}y_2 \end{aligned}\right\} \cdots (1)$$

We can graph solutions of (1):

$Y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}$ as two curves over the t-axis, one for each component of $Y(t)$

But we can also graph it as a single curve in the y_1y_2 plane (Parametric representation with parameter t)

Such a curve is called a Trajectory (or orbit or path) of (1).

The y_1y_2-plane is called the Phase plane.

If we fill the phase plane with trajectories of (1), we obtain Phase portrait of (1).

A phase portrait gives a good general qualitative impression of the entire family of solutions.

Ex:

$$Y' = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} Y$$

$\Rightarrow A = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} \Rightarrow$ characteristic polynomial of A: $(\lambda+3)^2 - 1 = 0$

$$\Rightarrow \lambda^2 + 6\lambda + 8 = 0$$

$$\Rightarrow (\lambda+4)(\lambda+2) = 0$$

$$\Rightarrow \lambda = -2, -4$$

For $\lambda = -2$, $AX = \lambda X \Rightarrow \left.\begin{aligned} -3x_1 + x_2 &= -2x_1 \\ x_1 - 3x_2 &= -2x_2 \end{aligned}\right\}$

$$\Rightarrow -x_1 + x_2 = 0 \Rightarrow X_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda = -4$, $AX = \lambda X \Rightarrow \left.\begin{aligned} -3x_1 + x_2 &= -4x_1 \\ x_1 - 3x_2 &= -4x_2 \end{aligned}\right\}$

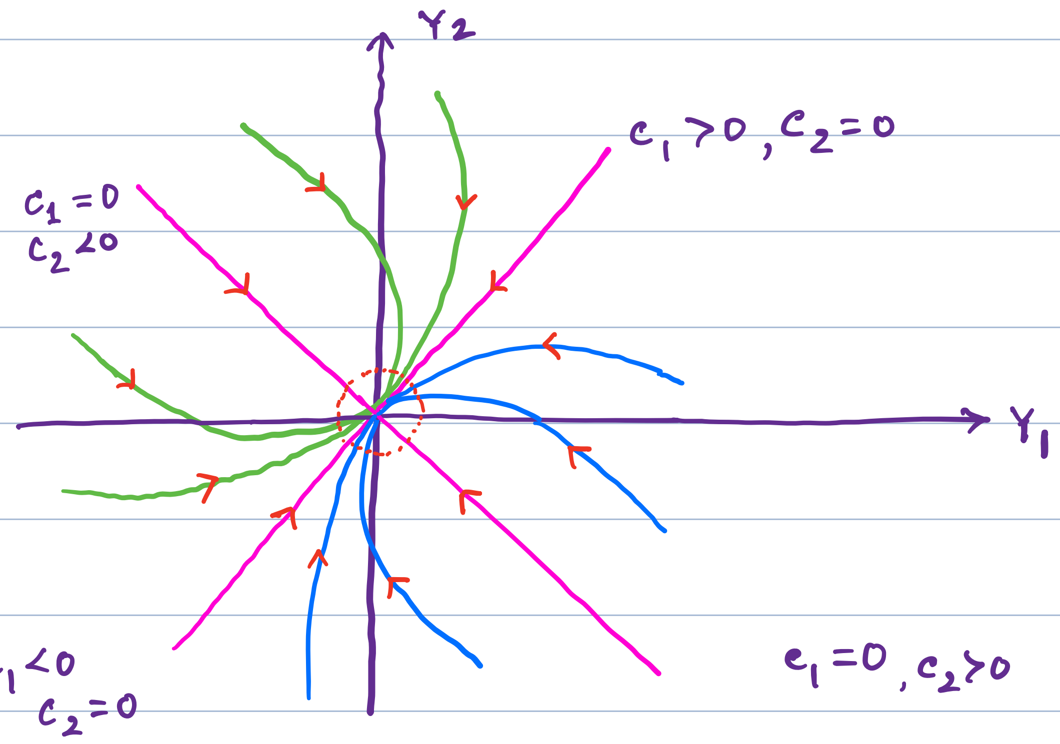
$$\Rightarrow x_1 + x_2 = 0 \Rightarrow X_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Then $Y = c_1 X_1 e^{\lambda_1 t} + c_2 X_2 e^{\lambda_2 t}$

$$= c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-4t}$$

If $c_1 = 0$, $y_1 = c_2 e^{-4t}$, $y_2 = -c_2 e^{-4t} \Rightarrow y_1 = -y_2$

If $c_2 = 0$, $y_1 = c_1 e^{-2t}$, $y_2 = c_1 e^{-2t} \Rightarrow y_1 = y_2$



Critical points:

A critical point is a point at which $Y' = 0$.
They are also called equilibrium solutions.
From the equation of the system

$$Y' = AY,$$

slope of the trajectories in the phase plane at a given point (y_1, y_2):

$$\frac{dy_2}{dy_1} = \frac{\frac{dy_2}{dt}}{\frac{dy_1}{dt}} = \frac{y_2'}{y_1'} = \frac{a_{21}y_1 + a_{22}y_2}{a_{11}y_1 + a_{12}y_2}$$

(At every point, there is a unique tangent direction $\frac{dy_2}{dy_1}$ of the trajectory passing through that point.)

At critical points, this ratio becomes undefined $\left(\frac{0}{0}\right)$.

There are five types of critical points:
improper nodes, proper nodes, saddle points,

centers and spiral points.

<u>Improper node:</u> An improper node is a critical point at which all trajectories, except two of them have the same limiting direction of the tangent. The two exceptional directions also have a limiting direction of the tangent which however is different.

<u>Ex ①:</u> $Y' = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} Y$

$(0,0)$ is a critical point

since $Y' = 0 \Rightarrow \left.\begin{matrix} -3y_1 + y_2 = 0 \\ y_1 - 3y_2 = 0 \end{matrix}\right\} \Rightarrow y_1 = 0, y_2 = 0$

We have calculated the general solution of the above system as: $Y = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-4t}$

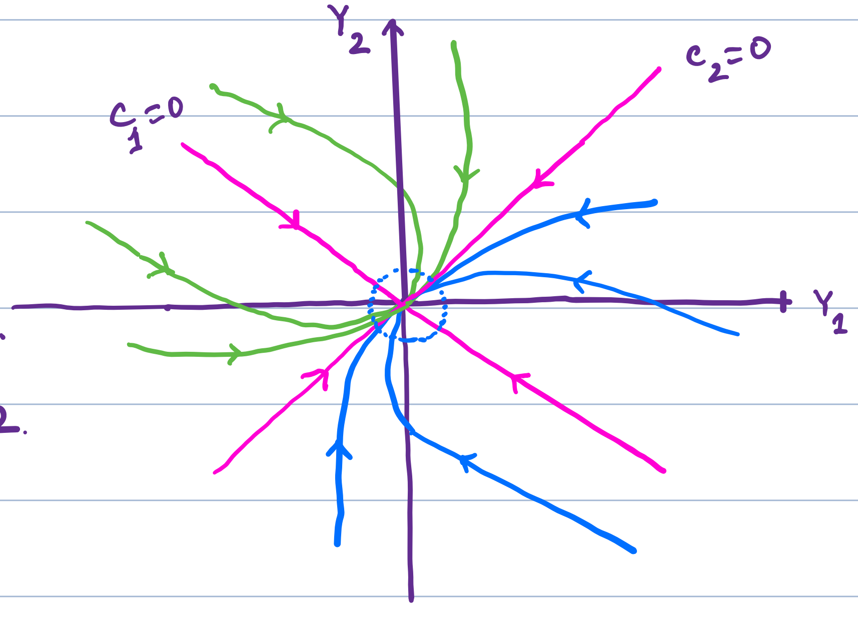
The common limiting direction at 0 is the direction of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ because e^{-4t} goes to zero faster than e^{-2t} as t increases to ∞.

The two exceptional limiting tangent directions are those of $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$(0,0)$ is an improper node.

eigen values are real and distinct and of same sign.

dimension of eigen space $= 2$.



Ex: Proper node:

Consider the system $Y' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} Y$

The characteristic equation is $(1-\lambda)^2 = 0 \Rightarrow \lambda = 1$

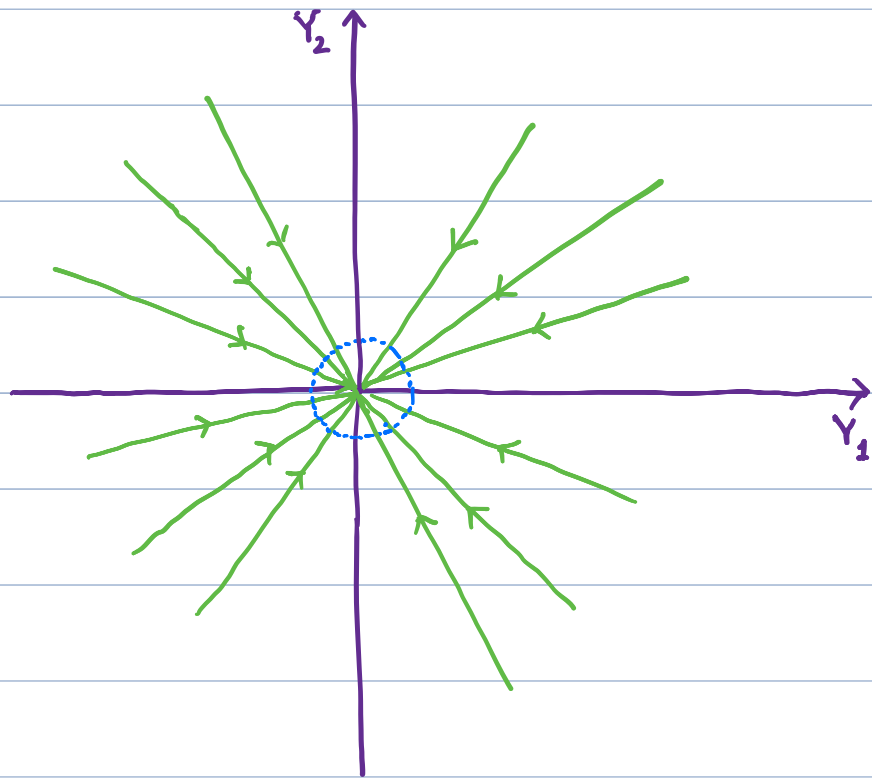
From $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 1 \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \left.\begin{matrix} x_1 = x_1 \\ x_2 = x_2 \end{matrix}\right\} \Rightarrow$ Two linearly independent eigenvectors are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

So, the general solution is: $Y = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^t$

A proper node is a critical point at which every trajectory has a definite limiting direction and for any given direction d, there is a trajectory having d as its limiting direction.

$Y_1 = c_1 e^t, \quad Y_2 = c_2 e^t$

$\Rightarrow \quad c_2 Y_2 = c_1 Y_1$



$(0,0)$ is a proper node.

Eigen values are real and equal.

(eigen space has dimension 2)

(It is also called a Star)

Ex: Saddle Point

Let $Y' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} Y$: The characteristic equation is $(1-\lambda)(-1-\lambda) = 0$

$\Rightarrow \lambda^2 - 1 = 0 \Rightarrow \lambda = 1, -1$, From $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow x_1 = x_1, -x_2 = x_2 \}$

From $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -1\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \left.\begin{matrix} x_1 = -x_1 \\ -x_2 = -x_2 \end{matrix}\right\} \Rightarrow$ Two linearly independent eigen vectors

are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. So, the general solution is $Y = c_1\begin{pmatrix} 1 \\ 0 \end{pmatrix}e^{t} + c_2\begin{pmatrix} 0 \\ 1 \end{pmatrix}e^{-t}$

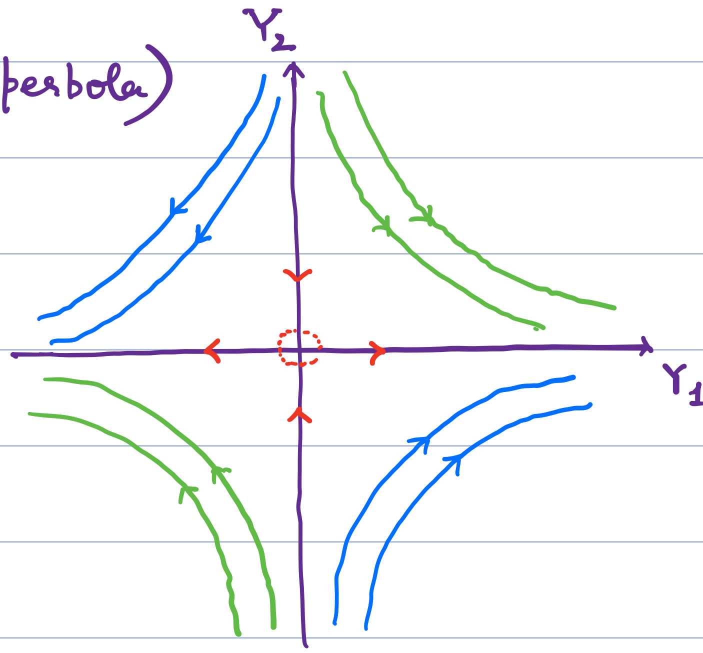
A saddle point is a critical point at which there are two incoming trajectories, two outgoing trajectories and all other trajectories in a neighborhood of the critical point bypass it.

$Y_1 = c_1 e^{t}, \; Y_2 = c_2 e^{-t}$

$\Rightarrow Y_1 Y_2 = c_1 c_2$ (rectangular hyperbola)

If $c_2 = 0$, $Y_2 = 0$, $Y_1 = c_1 e^{t} \left.\begin{matrix} \to \infty \text{ if } c_1 > 0 \\ \to -\infty \text{ if } c_1 < 0 \end{matrix}\right\}$

If $c_1 = 0$, $Y_1 = 0$, $Y_2 = c_2 e^{-t} \left.\begin{matrix} \to 0 \text{ if } c_2 > 0 \\ \to 0 \text{ if } c_2 < 0 \end{matrix}\right\}$



The eigen values are real and distinct and of opposite sign.

Ex: Center: $Y' = \begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix} Y$. The characteristic equation is:

$(-\lambda)(-\lambda)+4=0 \Rightarrow \lambda^2+4=0$

$\Rightarrow \lambda = \pm 2i$: From $\begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2i\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow x_2 = 2i x_1, -4x_1 = 2i x_2$

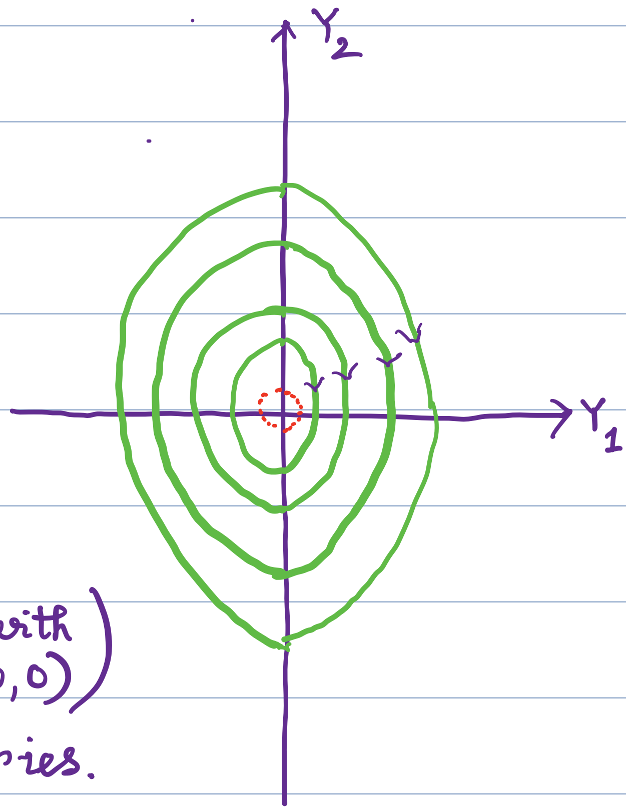
From $\begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -2i\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow x_2 = -2i x_1, -4x_1 = -2i x_2$

Two linearly eigenvectors are $\begin{pmatrix} 1 \\ 2i \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -2i \end{pmatrix}$

So, the general solution is $Y = c_1 \begin{pmatrix} 1 \\ 2i \end{pmatrix} e^{2it} + c_2 \begin{pmatrix} 1 \\ -2i \end{pmatrix} e^{-2it}$

- A center is a critical point that is enclosed by <u>infinitely many closed trajectories</u>.

$$\left.\begin{aligned} Y_1 &= c_1 e^{2it} + c_2 e^{-2it} \\ Y_2 &= 2i c_1 e^{2it} - 2i c_2 e^{-2it} \end{aligned}\right\}$$



$\Rightarrow Y_1' = Y_2$ and $Y_2' = -4Y_1$

$\Rightarrow 4Y_1 Y_1' = -Y_2 Y_2'$

$\Rightarrow \int 4Y_1 \, dY_1 = -\int Y_2 \, dY_2$

$\Rightarrow 2Y_1^2 + \frac{Y_2^2}{2} = C$

$\Rightarrow Y_1^2 + \frac{Y_2^2}{4} = C$ (family of ellipse with center $(0,0)$)

$(0,0)$ is the center of the trajectories.

(Eigen values are purely imaginary)

Spiral point:

A spiral point is a critical point P_0 about which the trajectories spiral approaching P_0 as $t \to \infty$ (or tracing these spirals in the opposite sense, away from P_0)

Ex: $Y' = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} Y$. The characteristic equation is

$$(-1-\lambda)(-1-\lambda) + 1 = 0$$

$$\Rightarrow \lambda^2 + 2\lambda + 2 = 0 \Rightarrow \lambda = \frac{-2 \pm \sqrt{4 - 4(2)}}{2} = -1 \pm i$$

From $\begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = (-1+i)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \left.\begin{array}{l} -x_1 + x_2 = (-1+i)x_1 \\ -x_1 - x_2 = (-1+i)x_2 \end{array}\right\} \Rightarrow \begin{array}{l} x_2 = ix_1 \\ -x_1 = ix_2 \end{array}$

From $\begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = (-1-i)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \left.\begin{array}{l} -x_1 + x_2 = (-1-i)x_1 \\ -x_1 - x_2 = (-1-i)x_2 \end{array}\right\} \Rightarrow \begin{array}{l} x_2 = -ix_1 \\ -x_1 = -ix_2 \end{array}$

Two linearly independent eigenvectors are $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$

So, the general solution is:

$$Y = c_1 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{(-1+i)t} + c_2 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{(-1-i)t}$$

$$\Rightarrow \left.\begin{array}{l} Y_1 = c_1 e^{(-1+i)t} + c_2 e^{(-1-i)t} \\ Y_2 = ic_1 e^{(-1+i)t} - ic_2 e^{(-1-i)t} \end{array}\right\}$$

$$\Rightarrow Y_1' = c_1(-1+i)e^{(-1+i)t} + c_2(-1-i)e^{(-1-i)t}$$

$$\Rightarrow Y_1' = (-c_1 e^{(-1+i)t} - c_2 e^{(-1-i)t}) + ic_1 e^{(-1+i)t} - ic_2 e^{(-1-i)t}$$

$\Rightarrow \; Y_1' = -Y_1 + Y_2 \quad \cdots\cdots\cdots (1)$

and $Y_2' = i(-1+i)c_1 e^{(-1+i)t} - i(-1-i)c_2 e^{(-1-i)t}$

$= (-c_1 e^{(-1+i)t} - c_2 e^{(-1-i)t}) - (ic_1 e^{(-1+i)t} - ic_2 e^{(-1-i)t})$

$Y_2' = -Y_1 - Y_2 \quad \cdots\cdots (2)$

Now $(1) \times Y_1 + (2) \times Y_2 \Rightarrow Y_1'Y_1 + Y_2'Y_2 = -Y_1^2 + Y_1Y_2 - Y_1Y_2 - Y_2^2$

$\Rightarrow Y_1'Y_1 + Y_2'Y_2 = -Y_1^2 - Y_2^2 \Rightarrow Y_1'Y + Y_2'Y_2 = -(Y_1^2 + Y_2^2)$

we now introduce polar coordinates r and t such that $r^2 = Y_1^2 + Y_2^2$ (ie. $Y_1 = r\cos t$, $Y_2 = r\sin t$)

$$\Rightarrow \quad 2r\frac{dr}{dt} = 2Y_1\frac{dY_1}{dt} + 2Y_2\frac{dY_2}{dt} = 2Y_1Y_1' + 2Y_2Y_2'$$

$$\Rightarrow 2r\frac{dr}{dt} = -2r^2 \Rightarrow \frac{dr}{dt} = -r$$

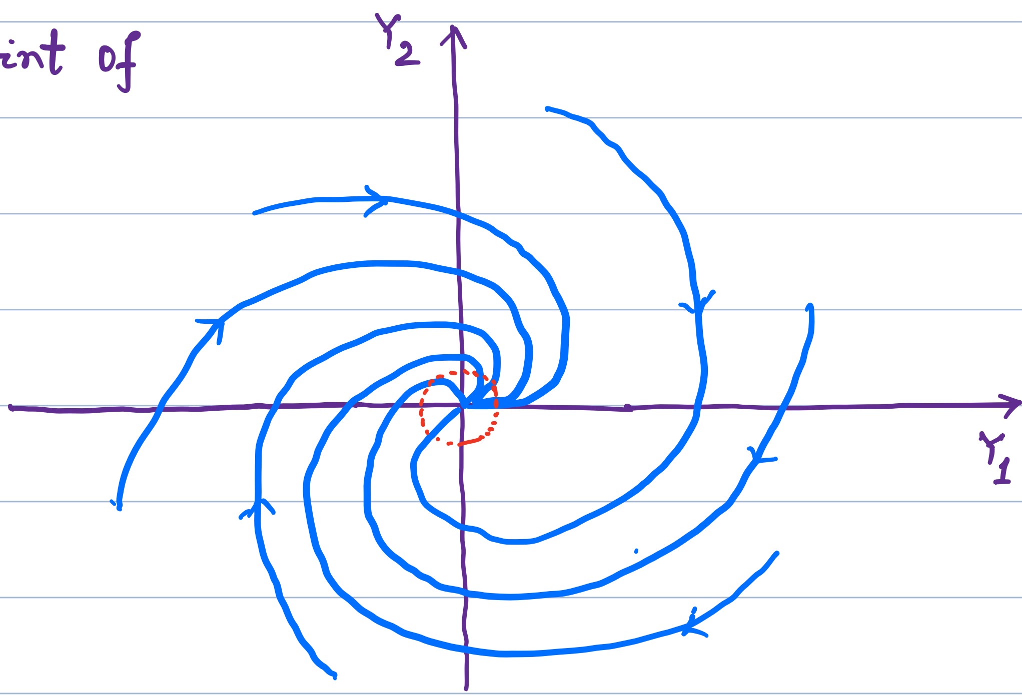
$$\Rightarrow \frac{dr}{r} = -dt \Rightarrow \ln|r| = -t + c_1$$

$$\Rightarrow \boxed{r = ce^{-t}} \quad (\text{where } c = e^{c_1})$$

For each real c, this is a spiral

(0,0) is a spiral point of the trajectories.

(Eigen values are complex but not purely imaginary).



Degenerate node:

Ex: $Y' = \begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix} Y$ Let $A = \begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}$

The characteristic equation is $(4-\lambda)(2-\lambda)+1=0$

$\Rightarrow 8-6\lambda+\lambda^2+1=0 \Rightarrow \lambda^2-6\lambda+9=0 \Rightarrow (\lambda-3)^2=0$

Thus $\lambda_1 = 3$ is an eigen value of the coefficient matrix with multiplicity 2

From the equation $\begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 3\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \Rightarrow \left.\begin{array}{l} 4x_1+x_2=3x_1 \\ -x_1+2x_2=3x_2 \end{array}\right\}$

$\Rightarrow \left.\begin{array}{l} x_1+x_2=0 \\ -x_1-x_2=0 \end{array}\right\}$ So, we get only one independent eigen vector $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

So, one of the solutions is of the form

$$Y_1 = X_1 e^{\lambda_1 t} = X_1 e^{3t} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{3t}$$

For the second solution we try

$Y_2 = t X_1 e^{\lambda_1 t} + u e^{\lambda_1 t}$ where u is a constant vector.

Then $Y_2' = X_1 e^{\lambda_1 t} + t\lambda_1 X_1 e^{\lambda_1 t} + \lambda_1 u e^{\lambda_1 t}$

Since $Y_2' = AY_2$, we have

$$x_1 e^{\lambda_1 t} + t\lambda_1 x_1 e^{\lambda_1 t} + \lambda_1 u e^{\lambda_1 t} = te^{\lambda_1 t} Ax_1 + e^{\lambda_1 t} Au$$

$$\Rightarrow x_1 e^{\lambda_1 t} + \cancel{t\lambda_1 x_1 e^{\lambda_1 t}} + \lambda_1 u e^{\lambda_1 t} = \cancel{t\lambda_1 x_1 e^{\lambda_1 t}} + e^{\lambda_1 t} Au$$

$$\Rightarrow x_1 e^{\lambda_1 t} + \lambda_1 u e^{\lambda_1 t} = Au e^{\lambda_1 t}$$

$$\Rightarrow x_1 + \lambda_1 u = Au$$

$$\Rightarrow (A - \lambda_1 I) u = x_1$$

$$\Rightarrow \begin{bmatrix} 4-3 & 1 \\ -1 & 2-3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

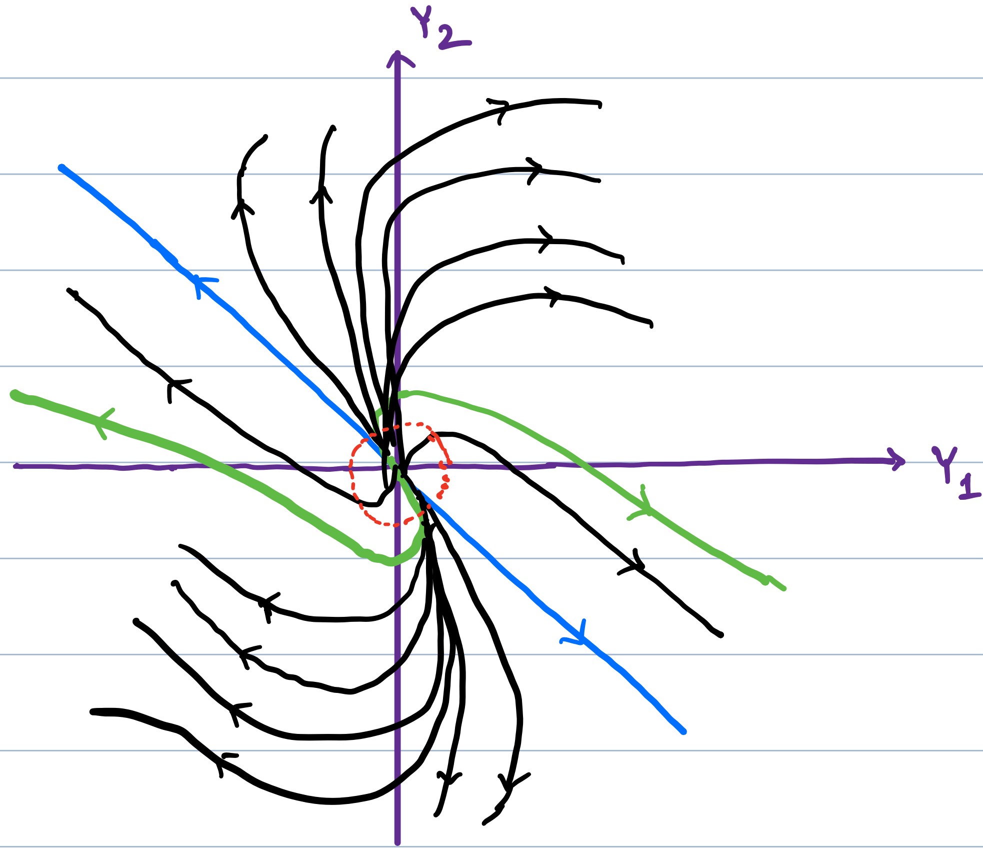
$$\Rightarrow \left.\begin{aligned} u_1 + u_2 &= 1 \\ -u_1 - u_2 &= -1 \end{aligned}\right\} \Rightarrow u_1 + u_2 = 1$$

So, a solution, linearly independent of $x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is $u = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

So, the general solution is:

$$Y = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{3t} + c_2 \left(t \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{3t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{3t} \right)$$

The critical point at the origin is called a degenerate node.



Note: When the matrix A is not diagonalizable and there is one linearly independent eigenvector x_1 corresponding to an eigenvalue λ_1, we may complete the fundamental system with solutions of the form:

$$Y_2 = (t\,x_1 + v_1)\,e^{\lambda_1 t}$$

$$Y_3 = \left(\frac{1}{2}t^2 x_1 + t\,v_1 + v_2\right)e^{\lambda_1 t}$$

$$Y_4 = \left(\frac{1}{3}t^3 x_1 + \frac{1}{2}t^2 v_1 + t\,v_2 + v_3\right)e^{\lambda_1 t}$$

$$\vdots$$